

Save Your Photo

Group 8

廖邦宸*, 林鈺承†, 張愛翎†, 洪渝喬†, 徐綉惠*

National Tsing Hua University, Hsinchu, Taiwan

*Dept. of Communications Engineering

Student ID: 113064542, 113064525

†Dept. of Electrical Engineering

Student ID: 113061537, 113061551, 113061533

Abstract—We present a generative framework designed to rectify suboptimal photographic compositions by adjusting camera angles post-capture. Addressing the high remediation cost of “bad photos,” our method leverages a frozen Stable Diffusion 1.5 backbone integrated with a Dual-ControlNet architecture to decouple appearance from structural guidance. We introduce a custom dataset comprising 698 paired images (synthetic and real) to support this task. Experimental results demonstrate that utilizing Depth Maps combined with an image-to-image pipeline achieves superior structural consistency, yielding an FID score of 102.88. Our approach effectively bridges the gap between amateur photography skills and professional aesthetics without requiring physical retakes.

I. INTRODUCTION

With the ubiquity of smartphones, photography has become an integral part of daily life. However, a significant gap remains between the aesthetic expectations of users and the actual photographic skills of the general public.

Consider a real-world scenario involving a high-cost travel destination: a couple travels to a remote location (e.g., the terminal station of a light rail transit) to capture a specific moment. If the resulting image suffers from poor composition or suboptimal perspective, the cost of remediation is extremely high. Unlike color correction or exposure adjustment, which can be easily handled by traditional image processing tools, fixing the camera angle or perspective typically requires a physical revisit—in this case, waiting for the next transit and spending hours retracing the journey.

This example highlights a critical limitation in current photography workflows: geometric attributes (such as viewpoint and composition) are traditionally considered “fixed” once the shutter is pressed. Our project aims to challenge this limitation by leveraging Generative AI to allow for post-capture perspective correction, thereby eliminating the need for physical retakes and significantly improving user experience.

Our framework builds upon Stable Diffusion 1.5 with a frozen U-Net to preserve generative quality. We integrate two fine-tuned ControlNet branches: one conditioned on Depth/Pose maps to enforce structural consistency, and



Fig. 1: Samples of our main experimental results. In each column, the left image shows the bad-angled input, while the right image shows the enhanced good-angled image generated by our model.

another conditioned on the source RGB image to preserve visual identity and color. The outputs of these control branches are aggregated and injected into the main U-Net decoder blocks to guide the image synthesis.

To train our model, we also propose a paired dataset consisting of 698 pairs of good-angled and bad-angled images, featuring various synthetic and real environments, characters, and poses. This dataset helps the model gener-

alize across different environments and character configurations, thereby addressing the problem of inconsistent or suboptimal camera angles in real-world images.

II. RELATED WORK

In this section, we review the foundational technologies and state-of-the-art methods that form the basis of our research. We first discuss DiffPortrait360, a recent advancement in 360-degree view synthesis which serves as a primary reference for our work. Subsequently, we introduce the Stable Diffusion framework and ControlNet, which are the core backbones of our proposed architecture.

A. DiffPortrait360: Consistent Portrait Diffusion

Generating consistent 360-degree views from a single portrait requires handling blind spots (e.g., the back of the head) and ensuring smooth transitions. DiffPortrait360 [1] addresses this by employing a frozen pre-trained Latent Diffusion Model (LDM) as the backbone, incorporating three auxiliary trainable modules for disentangled control:

- **Dual Appearance Module ($\mathcal{R}$):** To prevent artifacts caused by missing information in the single frontal view, this module extracts appearance features from both the reference image (I_{ref}) and an auxiliary back-view image (I_{back}). The I_{back} is synthesized by a specialized Back-view Generation Module ($\mathcal{F}$).
- **View-Control Module ($\mathcal{C}$):** This module is responsible for precise camera control. It derives camera pose conditions which are then rendered using an off-the-shelf 3D GAN, ensuring the generated content aligns with the desired geometric perspective.
- **View-Consistency Module ($\mathcal{V}$):** To solve the flickering and choppy transitions often seen in diffusion-based generation, this module integrates with a sequential training strategy. It enhances continuity between frames, allowing the model to produce locally continuous view sequences suitable for downstream tasks like NeRF reconstruction.

The overall architecture, illustrating the interaction between these modules and the frozen LDM, is shown in Fig. 2.

B. Stable Diffusion and Latent Diffusion Models

Stable Diffusion (SD), specifically version 1.5, serves as the generative backbone for both the reference method and our proposed approach. It is a prominent example of a Latent Diffusion Model (LDM) [2], which addresses the high computational costs of operating directly in pixel space.

As shown in Figure 3, the LDM architecture consists of two main phases:

- 1) **Perceptual Compression:** A pre-trained Variational Autoencoder (VAE) encodes the input image x into a lower-dimensional latent representation $z = \mathcal{E}(x)$. This latent space is perceptually equivalent to the image space but computationally more efficient.

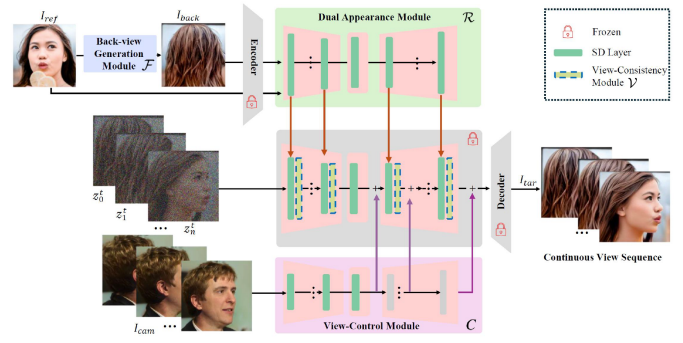


Fig. 2: The architecture of DiffPortrait360. It utilizes a frozen pre-trained LDM as the backbone and incorporates three trainable modules: the Dual Appearance Module ($\mathcal{R}$) for texture consistency, the View-Control Module ($\mathcal{C}$) for pose guidance, and the View-Consistency Module ($\mathcal{V}$) for smooth transitions [1].

- 2) **Latent Diffusion Process:** The diffusion process (adding and removing noise) occurs entirely within this latent space. A U-Net based denoising network ϵ_θ , equipped with cross-attention mechanisms, learns to denoise the latent variables z_t conditioned on inputs such as text or semantic maps.

Finally, the denoised latent vector is decoded back to the pixel space using the VAE decoder $\mathcal{D}$. This design allows for high-resolution synthesis with intricate details while maintaining tractable training and inference costs.

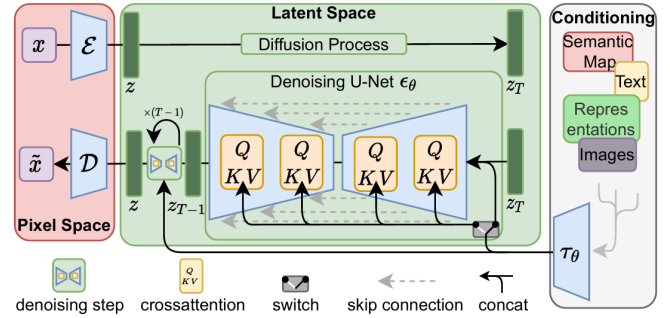


Fig. 3: The architecture of a Latent Diffusion Model (LDM) used in the Stable Diffusion 1.5 pipeline. The diffusion process operates in a compressed latent space facilitated by a VAE encoder $\mathcal{E}$ and decoder $\mathcal{D}$ [2].

C. ControlNet: Adding Conditional Control

While Stable Diffusion is powerful, controlling the spatial structure of the generated content (e.g., specific poses or depth maps) based solely on text prompts is difficult. ControlNet [3] provides a solution by introducing a neural network architecture that adds spatial conditioning to pre-trained diffusion models without fine-tuning the original weights.

As illustrated in Figure 4, ControlNet locks the parameters of the original production-ready model (gray blocks) and creates a trainable copy of the encoding layers (blue blocks). These two branches are connected via "zero convolution" layers, denoted as $\mathcal{Z}(\cdot; \cdot)$, which are 1×1 convolutions with both weights and biases initialized to zeros. To incorporate the external condition vector c , ControlNet utilizes two instances of zero convolutions with parameters Θ_{z1} and Θ_{z2} :

$$y_c = \mathcal{F}(x; \Theta) + \mathcal{Z}\left(\mathcal{F}(x + \mathcal{Z}(c; \Theta_{z1}); \Theta_c); \Theta_{z2}\right) \quad (1)$$

where y_c is the output, $\mathcal{F}$ represents the neural block, Θ are the frozen parameters, and Θ_c are the trainable parameters. In the first training step, since the zero convolution layers evaluate to zero, the equation simplifies to $y_c = \mathcal{F}(x; \Theta)$, ensuring that the model's initial behavior is identical to the original pre-trained backbone. During training, the trainable copy learns to incorporate the external conditions (e.g., Canny edges, depth maps, or human pose skeletons) and injects this information into the main model, effectively decoupling the learning of viewpoint transformations from the generative priors.

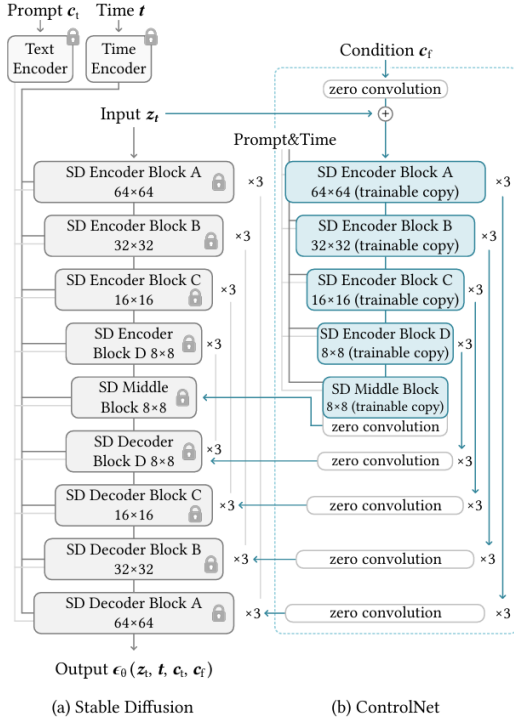


Fig. 4: Comparison of the original Stable Diffusion architecture (left) and the ControlNet architecture (right). ControlNet creates a trainable copy of the encoder blocks and connects them via zero convolution layers to inject conditional control [3].

III. PROPOSED DATASET

To train a model to correct camera angles, we require a paired dataset. The first image is a poorly angled image that serves as the input, while the second image is a well-angled image that serves as the ground truth. The model is expected to learn the transformation that maps the poorly angled image to the well-angled one. Below, we describe how we collect our proposed dataset.



Fig. 5: Samples from our proposed paired dataset. Each pair features good-angled (left) and bad-angled (right) image.

A. Synthetic Data Generation

Due to time and labor limitations, we were unable to capture all paired images in person. Therefore, we used the 3D software Blender to generate most of the dataset. We selected 13 wall types, 6 floor types, 40 different characters, and numerous lighting conditions with varying intensities and colors to construct the virtual scenes. Some assets were captured using our phones to obtain realistic textures, while others were downloaded from free online sources. The characters vary in gender, pose, and clothing. To generate an image pair, we randomly selected a wall, floor, lighting condition, and a character pose, and then rendered the same scene from two camera angles. One image was rendered from a top-down view with a random

camera tilt to simulate a poorly angled image, while the other was rendered from a bottom-up view with the character centered in the frame to serve as the well-angled ground-truth image. This process resulted in 600 pairs of synthetic data.

B. Real Data Collection

In addition to the synthetic data, real-world data is required to improve the model’s generalization to practical applications. We used iPhones to capture paired images of people with different poses and clothing. Due to practical limitations, the environments in the real images are less diverse than those in the synthetic dataset. As a result, we collected 98 pairs of real images featuring 9 different individuals with a variety of clothing styles.

In total, we collected 698 pairs of good and bad angle images, covering both synthetic and real environments with a diverse set of characters. Using this proposed dataset, we can train our model to perform camera angle correction on images. Samples of the dataset can be seen in Fig. 5.

IV. METHOD

For our proposed model architecture, we use Stable Diffusion 1.5 as the backbone, with two ControlNets to provide input conditions. Below, we describe each module of our model.

A. Stable Diffusion 1.5 Backbone

Following the architecture used in the reference work, we adopt Stable Diffusion 1.5 as the core generative backbone of our method. Stable Diffusion 1.5 is a latent diffusion model that operates in a compressed latent space rather than the pixel space. Specifically, the input image is first encoded into a low-dimensional latent representation using a pretrained variational autoencoder (VAE). The diffusion process and iterative denoising are then performed in this latent space by a UNet-based denoising network, significantly reducing computational cost while preserving high-quality visual details. Finally, the denoised latent representation is decoded back into the image space by the VAE decoder.

B. ControlNet Conditioning

To further improve controllability while avoiding re-training the entire diffusion backbone, we incorporate ControlNet into the Stable Diffusion framework. ControlNet augments a pretrained diffusion model by injecting additional conditional information through trainable residual branches, while keeping the original UNet parameters frozen. This design enables the model to leverage extra structural cues without sacrificing the generative prior learned by the base diffusion model.

In our approach, we employ two independently trainable ControlNet branches, each serving a distinct purpose. The first ControlNet is initialized with "tile" weight. It takes the RGB image captured from an undesirable viewpoint

as input. This branch encourages the model to preserve the original background, global layout, and fine-grained appearance details during generation. The second ControlNet is initialized with "depth" or "pose" weight. It’s conditioned on either a depth map, estimated via Depth Anything V2 [4], or a pose map, extracted using OpenPose [5]. These inputs provide explicit structural guidance on the human body geometry. This additional conditioning helps the model maintain consistent body proportions and spatial relationships while performing viewpoint transformation. By combining appearance-based and geometry-based conditioning through multiple ControlNet branches, our method effectively disentangles viewpoint adjustment from identity and background preservation, enabling controlled and realistic viewpoint transformation without re-training the full diffusion model.

C. Training and Inference Strategy

During training, the ground-truth images captured from desirable viewpoints are used as supervision for the diffusion model. Specifically, the target images are encoded into the latent space and serve as the denoising targets in the diffusion process. The input image with an undesirable viewpoint, together with its corresponding structural representation (either a depth map or a pose map), is fed into the two ControlNet branches as conditioning signals. The loss function during training is the MSE loss between the predicted noise and the ground truth noise. The training process and model structure is shown in Fig. 6.

Throughout training, the Stable Diffusion backbone is kept frozen to preserve its strong generative prior and generalization capability. In contrast, both ControlNet branches are fully fine-tuned, enabling the model to learn how to preserve scene appearance and identity while performing viewpoint transformation toward a visually improved perspective.

During inference, all inputs are images captured from undesirable viewpoints. We adopt an img2img [6] generation strategy, where the input image is first encoded into the latent space and partially noised, rather than generating images from pure random noise. With the guidance of the trained ControlNet branches, this formulation allows the model to better reconstruct the original scene and human appearance, while focusing the generation process on correcting the viewpoint.

V. EXPERIMENTS

A. Experimental Setup

We implemented our method using Stable Diffusion 1.5 as the generative backbone. To achieve structural control over the generated images, we integrated ControlNet to guide the synthesis process. Specifically, we utilized *OpenPose* and *Depth Anything V2* to extract pose and depth maps from the source images, respectively. These maps serve as conditional inputs to the ControlNet, ensuring that the generated output maintains the spatial structure

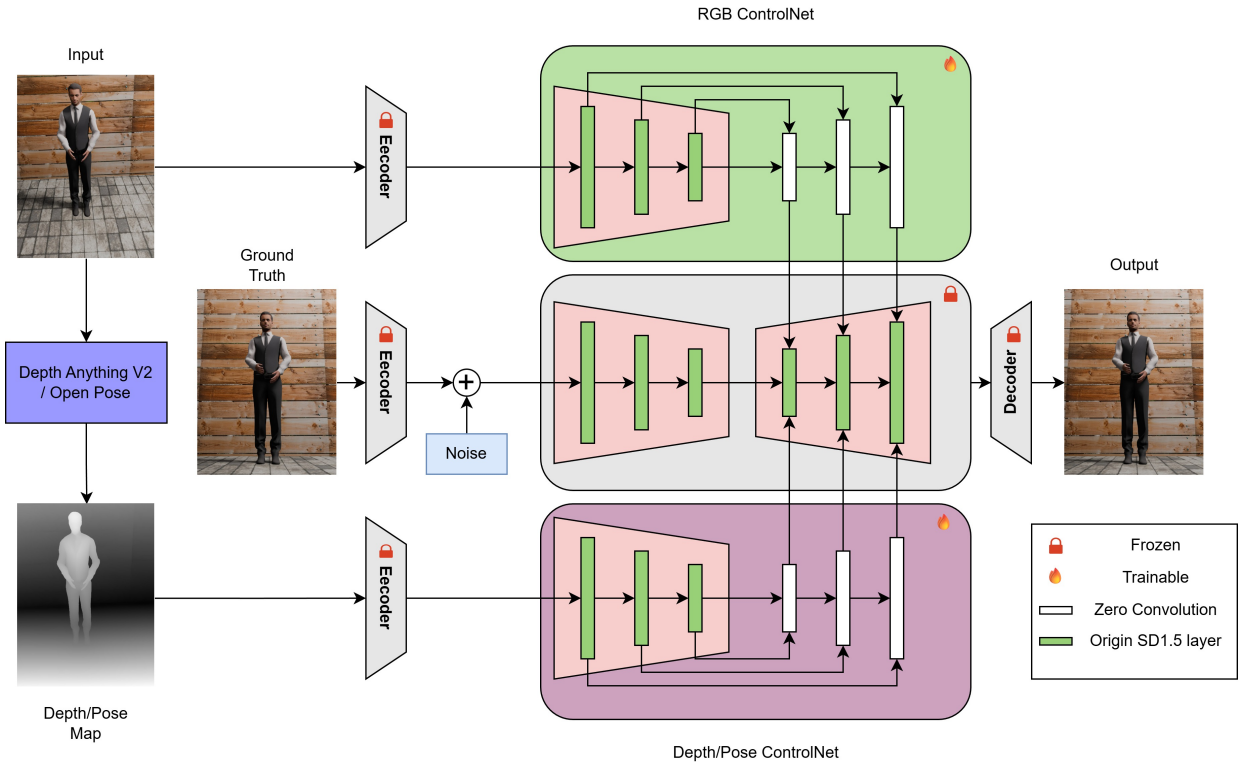


Fig. 6: Overall proposed model during training.

and semantic content of the original scene while correcting the camera angle. In the experiments below, we split our dataset into a 95% training set and a 5% testing set. All evaluations are performed on the testing set.

B. Quantitative Evaluation

We evaluated the quality of the generated images using the Fréchet Inception Distance (FID) metric, where a lower score indicates higher image quality and greater similarity to the real data distribution. The FID score is calculated between the prediction of the testing set and their ground truth. We conducted an ablation study to assess the impact of different ControlNet conditions (Pose Map vs. Depth Map) and the effectiveness of the image-to-image (img2img) pipeline.

TABLE I: Comparison of FID Scores across different ControlNet configurations. Lower FID indicates better image quality. The checkmark (✓) denotes the activation of the img2img pipeline. The red color denotes the changed method.

Method / Input Modality	img2img	FID ↓
RGB Only	×	113.32
RGB + Pose Map	×	108.98
RGB + Pose Map	✓	105.12
RGB + Depth Map	✓	102.88

The quantitative results are summarized in Table I. The baseline method using only RGB input yielded the highest

FID of 113.32. Incorporating structural guidance via a Pose Map improved the FID to 108.98. Activating the img2img pipeline further enhanced the quality, reducing the FID to 105.12 for the Pose Map configuration. The best performance was achieved by combining the Depth Map with the img2img pipeline, reaching the lowest FID score of 102.88. This indicates that depth information provides more robust structural guidance than pose estimation for this specific task.

C. Qualitative Analysis

In addition to quantitative metrics, we visually inspected the generated results to understand how different ControlNet configurations affect the output quality. Fig. 7 illustrates the visual differences using multiple test cases.

The first column shows the ground truth images. The second column displays results using the Pose Map. While the model captures human posture, background integration and detail consistency are occasionally limited. The last column demonstrates results using the Depth Map (our best method). The depth information provides more detailed spatial context, resulting in higher fidelity, better composition, and more natural integration between the subject and the background.

The second column, as shown in Fig. 8, displays the results generated without ControlNet. As observed, without explicit structural constraints, the model struggles to preserve the subject’s original pose and the scene’s



Fig. 7: Qualitative comparison of generated results. Left: Ground Truth (GT); middle: result using Pose Map; right: our best result, using Depth Map. The Depth Map approach generally yields superior structural consistency and visual quality.



Fig. 8: Visual ablation study demonstrating the importance of ControlNet. The left column shows the inputs, and the middle column shows the results when ControlNet is removed (relying only on img2img). And the right column shows our proposed method (RGB + Depth Map), which successfully corrects the angle while preserving the subject’s structure.

geometry. The generated images often suffer from significant hallucinations or fail to correct the perspective effectively. In contrast, the last column demonstrates the results using the Depth Map ControlNet. The depth condition effectively anchors the generation process, ensuring that the subject’s structure is preserved while successfully adjusting the camera angle.

VI. DISCUSSION

Despite the promising results in composition correction, our current model faces challenges in fidelity preservation and geometric consistency.

Firstly, as shown in Fig. 9, we observe that fine-grained details—such as specific facial characteristics and subtle hand gestures—may not be perfectly reconstructed. This limitation stems from the generative priors of Stable Diffusion, which tends to “hallucinate” plausible features rather than strictly copying pixels, especially when the subject occupies a small region of the latent space.

Secondly, a “flattening effect” is occasionally noticeable in the generated subjects, where the subject appears compressed along the projection axis during aggressive perspective transformations. An example of this geometric distortion is presented in Fig. 10.

The aforementioned issues could potentially be mitigated by increasing the size of the training dataset, as 698 pairs of images are generally insufficient for diffusion model training or fine-tuning. In addition to expanding the dataset, facial distortions may also be alleviated by providing extra conditional information that explicitly focuses on facial regions, enabling the model to better preserve facial structure during generation.



Fig. 9: Comparison between the input and generated result illustrating the loss of fine-grained facial fidelity.



Fig. 10: Comparison exhibiting geometric distortion where the generated subject appears unnaturally flattened.

VII. CONCLUSION

In this work, we presented a generative framework designed to remediate suboptimal photographic compositions by adjusting camera angles post-capture. By leveraging Stable Diffusion 1.5 combined with ControlNet, we demonstrated that structural guidance is essential for correcting perspective errors such as high-angle or low-angle distortions without requiring a physical revisit.

Our investigation yields the following key conclusions:

- **Optimal Configuration:** Through extensive ablation studies, we identified that Depth Map guidance combined with the image-to-image (img2img) pipeline offers the most robust performance. This configuration achieved the lowest FID score of 102.88, outperforming Pose-based methods and pure RGB baselines.
- **Structural Consistency:** While Pose Maps capture body language, they often fail to preserve the spatial relationship with the environment. Depth Maps provide superior geometric context, ensuring better subject-background integration during the angle adjustment process.
- **Dataset Contribution:** We introduced a custom dataset consisting of 698 paired images (600 synthetic scenes generated via Blender and 98 real-world samples). This dataset addresses the scarcity of paired data for camera angle correction tasks and serves as a foundation for future research.

In summary, our method successfully bridges the gap between casual photography and professional aesthetics. Looking forward, we propose several directions for future research. First, to address the occasional loss of high-frequency details (such as background textures), we plan to incorporate super-resolution modules. Second, we aim to extend this framework to video applications, which requires addressing temporal consistency to ensure stability across frames. Finally, we intend to expand our dataset with a larger proportion of real-world images to further enhance the model’s generalization capabilities in complex environments.

VIII. MEMBER CONTRIBUTION

廖邦宸: Related work surveying, Pre-experiments, Writing report

林鈺承: Synthetic dataset collection, Experiments, Writing report

張愛翎: Real dataset collection, Experiments, Writing report

洪渝喬: Main experiments, Writing report

徐綉惠: Experiments, Writing report

REFERENCES

- [1] Y. Gu, P. Tran, Y. Zheng, H. Xu, H. Li, A. Karmanov, and H. Li, “DiffPortrait360: Consistent Portrait Diffusion for 360 View Synthesis,” *arXiv preprint arXiv:2503.15667*, 2025.
- [2] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, “High-resolution image synthesis with latent diffusion models,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022, pp. 10684–10695.
- [3] L. Zhang, A. Rao, and M. Agrawala, “Adding conditional control to text-to-image diffusion models,” in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023, pp. 3836–3847.
- [4] Z. Cao, G. Hidalgo, T. Simon, S. Wei, and Y. Sheikh, “OpenPose: Realtime multi-person 2D pose estimation using Part Affinity Fields,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 1, pp. 172–186, 2019.
- [5] L. Yang, B. Kang, Z. Huang, Z. Zhao, X. Xu, J. Feng, and H. Zhao, “Depth Anything V2,” in *Proc. Neural Inf. Process. Syst. (NeurIPS)*, 2024.
- [6] C. Meng, Y. He, Y. Song, J. Song, J. Wu, J.-Y. Zhu, and S. Ermon, SDEdit: Guided image synthesis and editing with stochastic differential equations,” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022.